

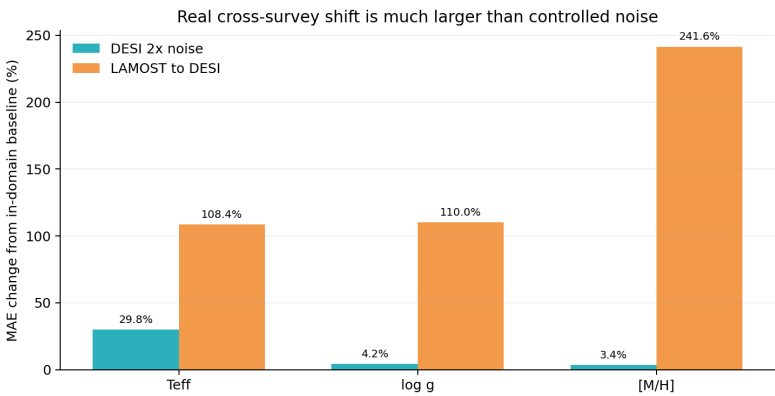
StellarShift-Bench v1.2.3

A prespecified quality-gated LAMOST DR2 to DESI DR1 reliability benchmark

1,088
model-held-out target stars

+108-242%
real-transfer MAE loss

28-60%
retraining recovery



The result

A LAMOST-trained model roughly doubles Teff/log g error and more than triples [M/H] error on DESI.

Source-calibrated nominal 90% coverage collapses to 68%, 75%, and 59%. Target recalibration restores near-nominal coverage.

CORAL harms [M/H] by 57.5%; labeled retraining improves all targets by 28%-60%.

Reliability layer	What was built	Why it matters
Data	Public LAMOST/APOGEE/DESI acquisition, exact hashes, object IDs	Reproducible and leakage-safe
Statistics	Per-star predictions, bootstrap CIs, paired method effects	Effect size, not p-value theater
Uncertainty	Split conformal under shift and target recalibration	Detects deployment overconfidence
Adaptation	Source-only, CORAL, retraining, 5-100 label budgets	Shows benefit, harm, and label efficiency
Robustness	Support, OOD, subgroups, model families, label errors	Tests where conclusions survive
Delivery	51 tests, deterministic release, CLI, manifests, report, notebook	Research-grade engineering

Scientific judgment

- The full effect mixes survey and population shift; a 918-star within-support audit still shows 32%-76% degradation.
- The common APOGEE scale controls label version, but is not independent truth.
- Hard bounds pass; isochrone consistency remains honestly unclaimed.

Role signal

- Research: falsifiable estimands, negative results, limitations.
- ML: calibration, OOD, adaptation, ablation, label efficiency.
- Engineering: typed contracts, public APIs, tests, provenance, archival release.

Deliverables: technical report, runnable results notebook, source release, evidence bundle, frozen acquisition manifest, and machine-readable per-star results.